

WORLDS OF TOMORROW

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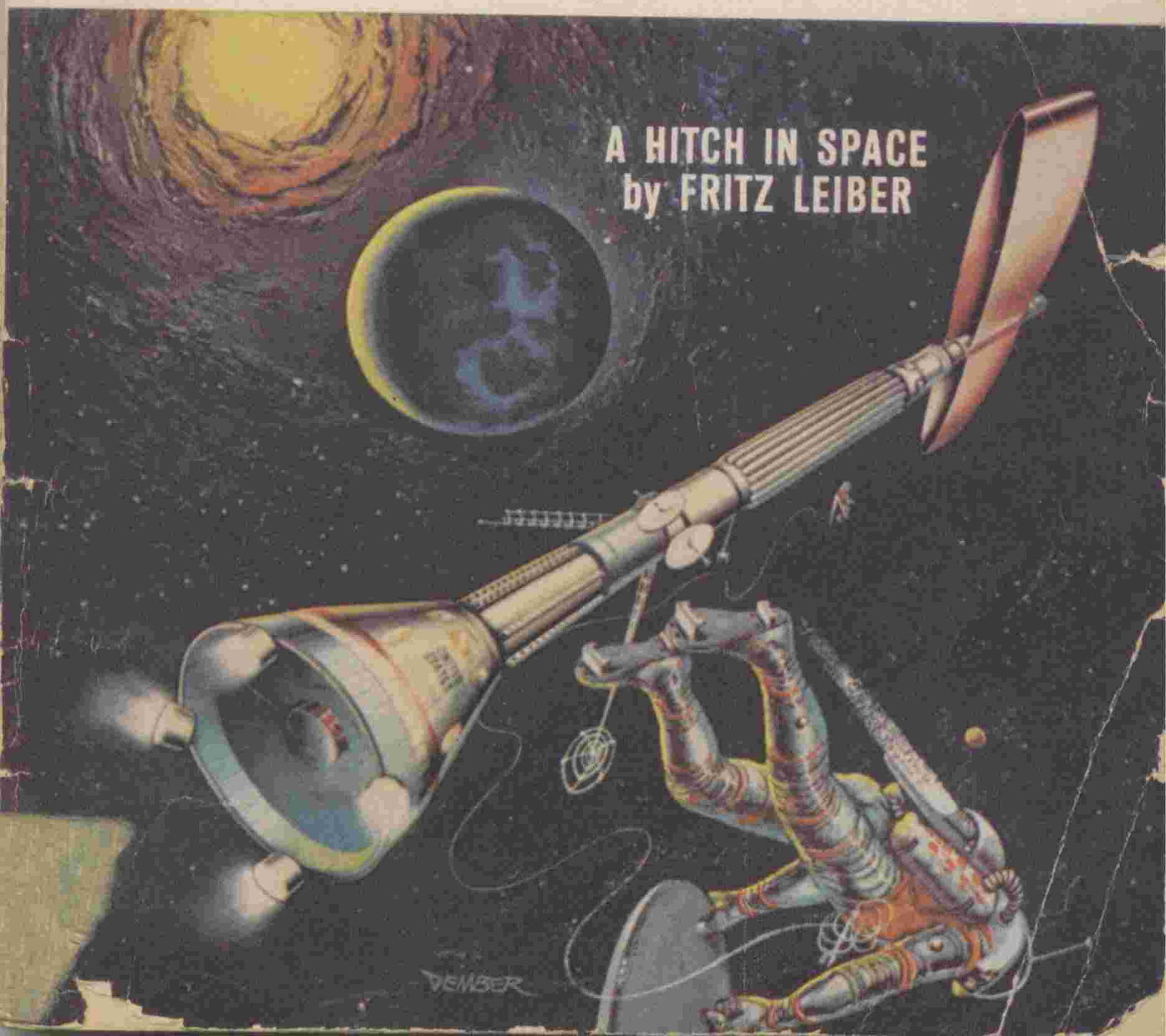
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MARSMEN
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THE NEW SCIENCE
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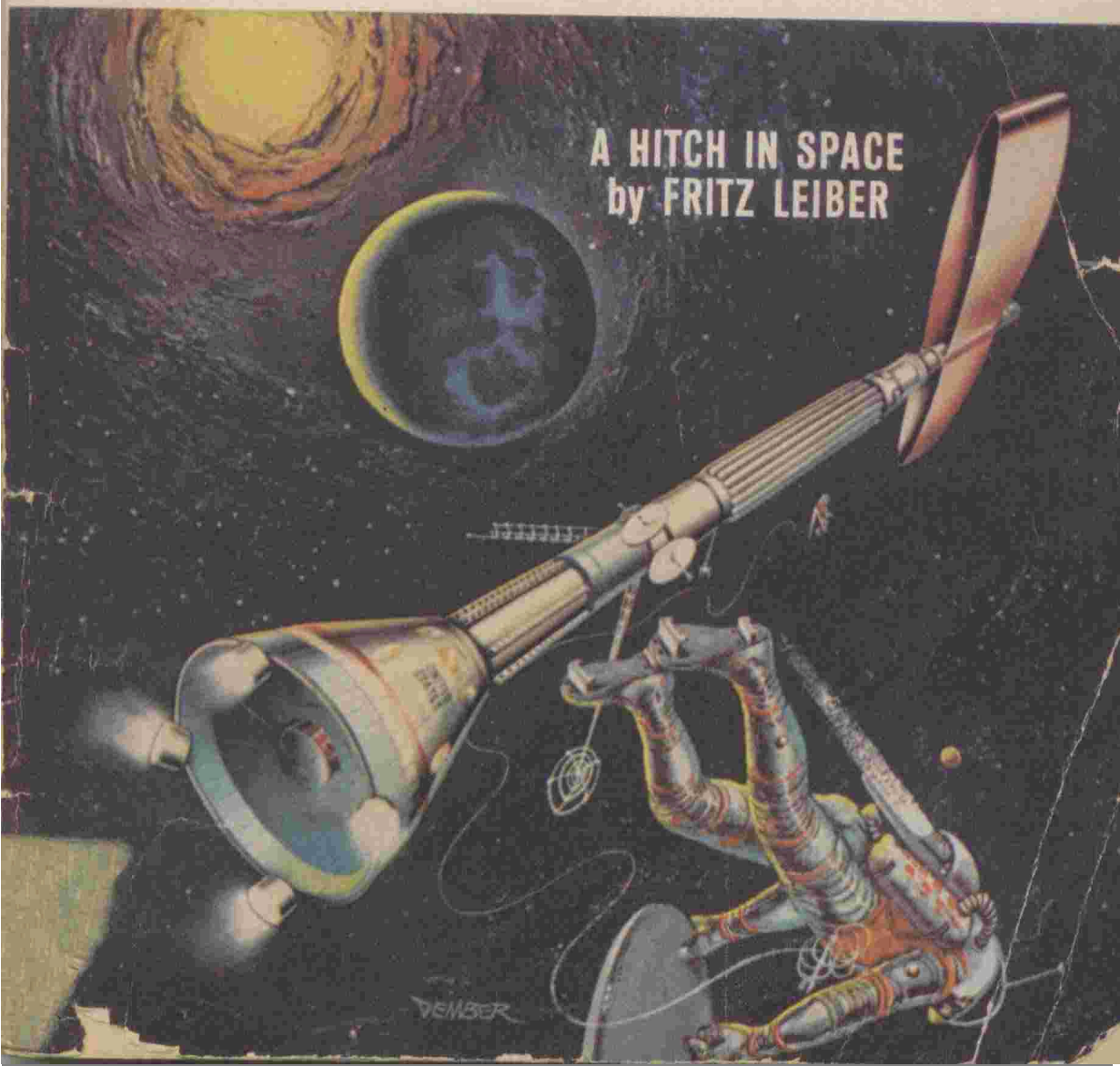
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***** START OF THE PROJECT GUTENBERG EBOOK THE NEW
SCIENCE OF SPACE SPEECH *****

The New Science of Space Speech

By Vincent H. Gaddis

How to talk to Martians, dolphins and creatures from the farthest stars—not tomorrow, but now!

A giant ear to listen to the whispers from infinity is being built at Sugar Grove, W. Va. This 600-foot radio telescope, largest ever designed, will cost \$100 million. When completed, its massive antenna, covering 6½ acres, will be trained on the mighty stellar mainland far beyond our solar system.

Astronomers believe that it will pick up cosmic impulses originating in stars from 60 to 80 light-years distant—seven times farther than America's largest existing radio telescope.

Meanwhile, a scientist in the Virgin Islands talks to a frisky dolphin. And the aquatic mammal replies, imitating the man's words with uncanny accuracy.

And at centers of learning in the United States and abroad scholars patiently work over mathematical charts and word lists, seeking formulas that will solve the problem of space speech.

These diverse activities are unified by a common purpose—to intercept and to interpret a possible message from outer space.

This signal across the vast void of the spaceways from intelligent but alien beings will be, perhaps, the most momentous event in human history. It could come tomorrow, or it may not be received for a century or more.

When it does come, man should be prepared to reply. This means we must devise some new method of communication that will transmit thoughts to non-human alien minds.

In awarding a contract for a space speech project, Dr. Dale W. Jenkins, chief of the National Space Administration's environment biology programs, stressed the great need for this knowledge.

“We have not yet determined whether there are any communications directed at earth from outer space,” he said. “If we do make contact, we will have to work out systems of understanding.”

This understanding is an all-important requisite as man reaches out toward the stars.

Understanding, however, will also have to be applied by man to himself when he joins the community of civilizations beyond.

Once interstellar intercourse is established, it will herald a new era in which man will have to recognize another species or form of life as intellectually his equal or more likely his superior. A recent psychological study of the possible effects of outer space contacts indicates that it will deflate human egoism with far-reaching consequences to his culture.

The problem of space speech is two-fold.

First, there are the techniques to be used in actual physical contact with other world inhabitants; second, the far more complex problem of exchanging concepts through the medium of radio communication.

Suppose you are a space explorer. You have landed on Mars or Venus and for the first time you are meeting intelligent creatures that are the products of a completely different line of evolution.

You possess five relatively well-developed senses. If the beings are not hostile, you must first determine if they have the same senses, only some part of them, or additional senses that man does not have.

For example, they may have a sense similar to extra-sensory perception and communicate with each other through telepathy. If you can exchange thoughts with them, that is fine. If you cannot tune in on their mental wavelengths, you're in trouble.

The sense of smell is practically limited to attractive perfumes and repulsive odors. Taste has the same limitations. Touch has been used for communication between humans, as in teaching the blind and deaf, but it requires physical contact (certainly a risky act when meeting strangers) and is limited to elementary concepts at best.

The only practical senses—of those which we humans possess, at any rate—for direct communication are sight and hearing.

If our Martians or Venusians have these senses—and if their reasoning processes are similar to those of humans—then communication could probably be established in the same manner with which we teach our children.

You could use “sign language.” You could point to your mouth and move your jaws to indicate you thought refreshments should be served. You could point to their head or heads (if they had them) and then at your own head and say “head.” With time and patience, a basis for communication could be established.

But suppose their methods of communication are entirely different. Suppose they use antennae, like ants, or gyrations, like bees.

Dr. Karl von Frisch, the German zoologist, discovered that when a bee locates a rich source of nectar, she returns to the hive and performs a dance. The number of times she turns reveals the distance, and her position in relation to the sun and the hive gives the direction.

This “breakthrough” into subhuman communication required controlled and sustained observation. It will have to be the necessary procedure if man encounters creatures with similar characteristics with his present knowledge.

Von Frisch’s discovery was a one-way avenue of understanding. But if the ants and the bees were much larger and more intelligent, we can assume that a demonstrative style of language could be devised for mutual communication.

To our scientists it is obvious that before our spacemen confront alien beings on a distant planet, we must learn the fundamentals of developing communication with a non-human but intelligent species right here on earth. And this is now in progress with “Project Dolphin.”

Bottle-nosed dolphins are not fish, but aquatic mammals. Often, but inaccurately, called porpoises, they are well known as clever, frolicsome entertainers at marineland exhibits.

Dolphins are by far the most intelligent animals other than man, and their brain power in some respects may even be superior to man’s. The dolphin brain is 40 per cent larger than the human, although smaller in proportion to

body weight, and the cerebral cortex—the layer of gray matter that originates rational thought—is just as complicated.

Dr. John C. Lilly, a neurophysiologist and a noted authority on the mammal, is in charge of the project. The research is principally being conducted at the Communications Research Institute of Charlotte Amalie, located at the U.S. Navy base on St. Thomas, Virgin Islands.

Dr. Lilly is working under a contract awarded in 1962 by the National Aeronautics and Space Administration. The contract is for basic scientific research “on the feasibility and methodology for establishing communications between man and other species.”

Dolphins have a complex vocal language. They talk to each other with sharp, high-pitched whistles and they talk almost continuously. Dr. Lilly has determined that the dolphin distress call is “an undulating sound,” with a rasping noise made periodically for range-finding.

Interpreting the dolphin vocabulary will not be easy since the creature emits heavy breathing sounds and there are other masking noises.

In experiments with ESB (electric stimulation of the brain), Dr. Lilly located the portion of the dolphin brain that created a feeling of pleasure. The dolphin almost immediately learned how to turn on a switch producing the current. For comparison, in similar tests it was found that monkeys required 300 or more tries before they attained their ability.

One day the electrical device broke down. The dolphin, annoyed at losing his pleasurable sensation, began making a series of sounds in imitation of the laboratory equipment. Dr. Lilly made a tape recording of these sounds.

Later the doctor played back the recording and in order to more distinctly hear the sounds he decided to run the tape at one quarter its normal speed.

It was then that Dr. Lilly made an astonishing discovery.

With exaggerated slowness, he listened to his own voice on the tape announcing the footage—“three, two, three”—and the dolphin immediately and clearly repeated the words in high-pitched whistles. Other tape recordings of what had seemed to be an unintelligible series of squawks and quacks, when played at half or quarter speed with the sound volume lowered, confirmed the discovery.

The dolphins were not only distinctly imitating the human words they heard, but were compressing their mimicry as to time. They were talking at

a rate eight times faster than humans.

One dolphin, Dr. Lilly recalls, “mimicked my speaking voice so well that my wife laughed out loud, and he copied her laughter.”

When one of the doctor’s assistants who had a southern drawl talked to one dolphin, the animal’s voice came back in clear imitation ... complete with the southern accent.

The next step—and it’s a big one—is to learn the dolphin language. The high-pitched, high-speed chatter must be broken down into definite meanings.

Dr. Frank D. Drake, director of Project Ozma (the recent attempt to receive possible messages by radio telescope), considers the dolphin language study to be of great importance.

He says the project “needs the skills of the radio astronomer in extracting signals from noise, and then the work of the linguist, and, perhaps, the cryptographer. It could well be, if the dolphin studies are correct, that we have right here on earth another intelligent race that is even more alien than some we might encounter in space.”

Second, there is the problem of interpreting and transmitting information through radio communication.

In April, 1960, Project Ozma was launched. The 85-foot radio telescope of the National Radio Astronomy Observatory at Green Bank, W. Va., was focused by government scientists on two stars in an attempt to pick up artificially produced signals.

The stars were Tau Ceti and Epsilon Eridani, 11 light-years or about 66 trillion miles away. They were chosen because they were similar to our own sun in size and rate of rotation.

The frequency at which natural hydrogen emits radio energy in space is 1,420 megacycles, and thus it is a universal constant. Dr. Drake tuned the receiver on both sides of this band.

Day after day the impulses were transmitted to a pen that traced erratic lines on a moving paper roll. But no repetitive pattern appeared that would indicate deliberate signals.

Early in 1961 it was announced that Project Ozma was being suspended. It is expected to be resumed when the new 600-foot radio telescope is completed.

The failure of Project Ozma to receive a message during a few months in operation is no surprise. In fact, it would not be a surprise if no signals were received during daily operation for a millennium.

There are known to be at least 100 quintillion stars. Focusing on one random star in the hope it has a planet having intelligent life beaming signals in our direction is like trying to find a specific drop of water in the ocean.

When a reporter during Project Ozma asked if there was any word from our remote fellow creatures, one scientist told him to come back in 10,000 years.

Yet certain factors may improve these chances. Advanced beings might periodically check the solar systems nearest them to see if they have company. It is not unreasonable to suppose that there is regular cosmic conversation between greatly developed cultures, and if we could detect a channel we might be able to plug in on the party line.

We can only hope, however, that they are using a method we can detect. Man has only recently emerged from savagery and is only beginning to look beyond his little world. To the cosmic callers, our most advanced equipment might be as primitive as smoke signals are to us.

Again, we might be trying to contact beings so entirely different from us that we would have no common ground upon which to build understanding. They might not even respond as we do to the same stimuli. Their appearance, evolution, structure, environment and thinking processes could even be beyond the limits of our imaginations.

But a signal could come—an impulse from out of the boundless abyss telling us we are not alone. What would be the nature of this message? And how could we reply?

Assuming that our senders are using radio wavelengths and have enough similarity to us for mutual understanding, we would first have to isolate the signals from the hash of natural static.

Next, we would have to “crack the code.” The usual cryptographic techniques, which depend on some basic knowledge of the language and

letter frequencies, would not be adequate. We can only hope that the callers give us some clues.

Scientists expect any messages received will be mathematical in nature, since mathematical principles may be regarded as universal constants.

The message might be a simple numeral progression or the numbers of a constant, such as the wave length of the hydrogen atom or the speed of light.

They might send *pi*, for example, the ratio of the circumference of a circle to its diameter. It's a non-stop number, but we would understand if it was worked out to six or eight decimal places. "*Pi* from the sky" would be the story of the ages.

Once we had received this signal for recognition and replied in equally simple terms would come the real problem of interpreting or devising a means for transmitting speed.

Hans Fruedenthal, professor of mathematics at the University of Utrecht (Netherlands), has devised a system he calls Lincos (meaning "Lingua Cosmica" or "Cosmic Language"). It consists of teaching the meaning of certain sounds by using numbers. The numbers would be signified by "dots" or "beeps"; the sounds by radio signals of various frequencies and lengths. To illustrate the method, let us assume that the sound "bloop" stands for "equal." Three dots would be sent, then bloop, then three dots. This would be repeated with other numbers until the listeners associated the sound with equal numbers.

The concept of "less than" would similarly be sent by several dots, another sound (like "tweet"), followed by a greater number of dots. The reverse—like a greater number of dots, another sound, and a lesser number of dots—would signify "greater than." Once these concepts were understood, the operative signs like add, subtract, etc. could be taught. Thus a mathematical vocabulary would be established.

Next would come transmitting the length of our basic time unit. The Fruedenthal system would send, say, a four-second dash, followed by the Lincos sound for "second," then four dots. Using different dash lengths with corresponding dots and the same sound, it is assumed that the

recipients would observe that the length of the dash was proportional to the number of dots.

Time concepts (including universal constants) would lead to teaching units of physical length.

Upon this foundation of mathematics, time and dimensions, Lincos develops an ingenious and extensive language for a detailed description of earth, its inhabitants and our culture.

Lincos, of course, assumes that the listeners are capable of understanding our mathematical concepts and that their reasoning processes are similar to ours. It illustrates one great fundamental difficulty in alien communication: whatever system we use, it has to be devised within the limitations of our one-planet knowledge and experience.

The basic principle of association (that is relating numbers to sounds to teach meanings) can be used in other systems. Some form of association, probably beginning with objects and sounds, will be necessary to teach dolphins a human language.

One other fundamental means of communication is being considered by scientists. This is the use of geometrical designs or symbols which would then evolve into pictures. It would be most practical in interplanetary communication.

A picture, as the Chinese say, is worth a thousand words.

In interstellar communication, geometrical figures could possibly be signified by numbers. Thus the pi ratio would denote a circle, three equal successive numbers an equilateral triangle, four equal numbers a square, and so on.

From this elementary basis, a method of translating sounds into drawings could be developed. This might take the form of having electrical circuits attached to pens or tiny lights respond to various sounds, thus transcribing the pictures to paper or film.

The correct interpretation of whatever messages we receive will be of extreme importance. An error could be disastrous.

We need only recall the difficulties we have had in translating early records of our own species to know that interpreting the signals of otherworld beings may be very difficult. Egyptian hieroglyphics were given

many translations that contradicted each other before the Rosetta Stone was found. In one example, there were 12 different translations.

Should this problem develop, we can only hope that the other-worlders are friendly, tolerant and patient.

Then there is the time factor.

If, during Project Ozma, a signal had been received and a reply sent, it would have been 22 years before we knew whether our answer had been received. A reply to a message from 80 light-years away received by the new radio telescope being built would take 160 years for confirmation.

Living languages are fluid. As new words are coined, others become obsolete. Definitions change with passing years.

King George I of England, upon inspecting Sir Christopher Wren's masterpiece, St. Paul's Cathedral, told the famous architect that his creation was "amusing, awful and artificial." Sir Christopher was delighted with the royal compliments.

Three centuries ago amusing meant amazing, awful meant awe-inspiring, and artificial meant artistic.

With time as dimension in universal communication, we would have to choose our words with care.

The accelerated scientific progress of recent years will doubtless continue, with new ways and means of cosmic communication being developed. Radio astronomy itself is barely three decades old. Revolutionary techniques in interstellar contacts may be just around the corner.

Has radio communication with alien beings already occurred? This is a startling possibility.

On August 22, 1924, the planet Mars approached to within thirty-four and a half million miles from the earth. Radio broadcasting stations were silenced and scientists listened for a possible message from across space.

At the suggestion of the late Dr. David Todd, professor emeritus of astronomy at Amherst College, the U.S. Government through diplomatic channels requested that all countries with high-power transmitters silence

their stations for five minutes every hour from 11:50 p.m. August 21 to 11:50 p.m. August 23.

Station WOR, Newark, N.J., reported receiving a word translated as “Eunza.” Other stations announced receiving strange signals.

Twenty-three years later, in 1947, Gene Darling, an early “ham” operator and General Electric Co. employee in Schenectady, N. Y., said he and an assistant had failed to turn off a test transmitter. “It kept on sending out automatic code signals,” he said, “and fearing criticism, we never told of our mistake.”

But something else happened during this 1924 test period of silence that remains a mystery today.

C. Francis Jenkins, of Washington, D. C., had only recently invented a radio photo message continuous-transmission machine. He was asked by Dr. Todd to take a record of any signals received during the periods of silence.

The recording device was attached to a receiver adjusted to the 6,000 meter wave length. Incoming signals caused flashes of light, which were printed on the film by an instrument passing over its surface from side to side. The film was in a roll, 30 feet long and six inches wide, and it was slowly unwound by clockwork under the instrument and light bulb which responded to transmitted sounds.

When the film was developed, it disclosed a fairly regular arrangement of dots and dashes along one side, but on the other side, at almost evenly spaced intervals, were curiously jumbled groups each taking the form of a man’s face.

Scientists at the radio division of the National Bureau of Standards and military code experts examined the film and admitted it was a freak that they couldn’t explain.

“The film of faces is a permanent record that can be studied,” Dr. Todd said, “and who knows just what these signals may have been?”

There have been other incidents. Marconi, the father of wireless, heard strange signals in 1921. And in 1928 Prof. A. M. Low, famous English scientist, listened to a “mysterious series of dots and dashes.”

Ham radio operators have occasionally reported curious stories. In QST, official organ of the International Amateur Radio Union, July, 1950, issue,

Byron Goodman, assistant technical editor of the magazine, tells of a ham receiving strange signals.

Certain unexplainable “echoes” were heard by scientists in 1927, and again in 1928 and 1934 while they were experimenting with the capabilities of radio. The Danish scientist, Hals, and two Scandinavian experimenters, Størmer and Petersen, received echoes from 280,000 to 2,800,000 miles from the earth.

Dr. Arthur C. Clarke reported that in a series of tests in Holland radio echoes of eight seconds delay (corresponding to a reflector at a distance of 744,000 miles) were obtained repeatedly in 1946.

What is the explanation?

Dr. Ronald N. Bracewell, professor of electrical engineering at Stanford University and co-author with J. L. Pawsey of a standard textbook (*Radio Astronomy*, Oxford University Press, 1955), has a theory. He suggests that some of these echoes may have come from a satellite in orbit around our sun.

If highly advanced beings have achieved space travel, placing a satellite in a solar system would be more practical than beaming radio signals continuously at thousands of stars for thousands of years.

Dr. Bracewell suggests that the experimental broadcasts included trigger signals that caused the satellite to respond with echoes. If the satellite’s reply was repeated by man, the satellite would probably release its store of information.

If man does make contact with a superior alien civilization, what will happen?

Recently the Brookings Institution released a report on this question. The study was made for the National Aeronautics and Space Administration at a cost of \$96,000.

If intelligent life is discovered on other worlds, the report warned, the stability of earth’s civilization will be threatened. It recommended a psychological preparation of human beings prior to the discovery.

“While the discovery of intelligent life in other parts of the universe is not likely in the immediate future,” the report said, “it could nevertheless

happen at any time.”

This is the lesson of history: When a culture is faced with a superior culture, it either disintegrates or is changed drastically.

Japan, when it was opened to the outside world, succeeded in adjusting to the new conditions. The Aztec culture collapsed.

Our beliefs, institutions and culture have been based on the premise that man is the most intelligent of creatures. Would we be able to assume a subordinate role?

Perhaps Dr. Otto Struve, the noted astronomer, was thinking about this when newsmen were interviewing him about Project Ozma. “I’m not so sure we should even answer if we did receive such signals,” he said.

Psychological preparation will certainly be needed.

Dr. Harlow Shapley, the Harvard professor emeritus of astronomy, after allowing for all elements of chance among the known stars, conservatively estimates that there should be a million planets with life-producing elements and conditions.

In all the vastness of space and eons of time, there must be intelligent life in myriad forms seeking other intelligent life for interstellar companionship.

When the signal comes, man will answer.

END

Transcriber’s Note

1. This story appeared in the August 1963 issue of *Worlds of Tomorrow*.
2. The original publication used "earth" to refer to the planet "Earth" consistently and "man" to refer to the collective "Man". Retained.

*** END OF THE PROJECT GUTENBERG EBOOK THE NEW
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